

# Causal Inference

## Lecture 5

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based on Prof. Arvid Sjölander's slides

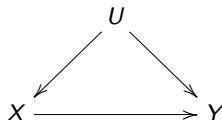
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- 1 IV - Introduction
- 2 Origins and overview
- 3 Hypothesis testing
- 4 Verifying the IV assumptions
- 5 The randomized trial with non-compliance
- 6 Effect estimation
  - The Two-Stage estimation method

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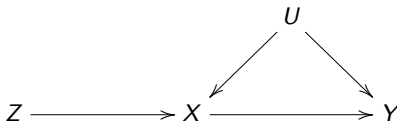
The examined problem:



- Simple causal structure where  $U$  confounds the effect of  $X$  on  $Y$ .
- Our goal is to identify and estimate the causal effect of  $X$  on  $Y$ .
- We assume that  $U$  is unmeasured (unobserved).

# Identification and Estimation of the causal effect $X \rightarrow Y$ : using IV

- When  $U$  is unmeasured, one can still identify the causal effect of interest by using a special variable  $Z$ .
- Such a variable is associated monotonically with the exposure  $X$  (relevance), affects the outcome  $Y$  only through the exposure (exclusion), and is not confounded with the outcome (exogeneity).
- These three properties define a valid instrumental variable (IV). Such variables allow for the identification of causal effects in the presence of unmeasured confounders  $U$ .



DAG of a valid instrumental variable  $Z$

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- **1928:** *Philip G. Wright* introduces instrumental variables in "The Tariff on Animal and Vegetable Oils."
- Aimed to estimate the causal effect of tariffs on market prices.
- First application of IVs to handle simultaneity bias.
- **1940s-1950s:** Formalization of IVs by economists such as *Ragnar Frisch* and *Trygve Haavelmo*.
- Recognized as essential tools for dealing with endogeneity in simultaneous equation models.
- **1970s-1980s:** Generalization of IV methods.
- Development of *Two-Stage Least Squares (2SLS)* by *Henri Theil* and others.
- IVs become a cornerstone for empirical research in social sciences.

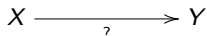
- 1990s-2000s: Expanded use in fields like medicine, political science, and epidemiology.
- Natural experiments and randomized trials emphasize IVs' importance.
- *Angrist and Krueger (1991)* popularized IVs in studying causal effects using natural experiments.
- Example: The impact of education on earnings using birth dates as instruments.
- In a series of papers in 1990's, Angrist, Imbens and Rubin connected IV to the potential outcomes framework in causal inference; Angrist, J. D., Imbens, G. W., & Rubin, D. B. (1996).



- Mendelian randomization is a method whose principles were introduced originally by Katan (1986) in a strictly medical context.
- Youngman et al. (2000) introduced this method in the context of epidemiological studies and also coined the aforementioned term.
- Mendelian randomization is a method that uses genotypes as IVs to estimate the causal effect of phenotype on disease-related outcomes.
- The population distribution of genetic variants is assumed to be independent of behavioral and environmental factors that usually confound the effect of exposure on the outcome.
- The process governing the distribution of genetic variants in the population resembles the randomization mechanism in RCTs. Namely, the genotype distribution constitutes a form of natural experiment that allows to identify the causal effect of exposure on the outcome.

# The aim of epidemiological research

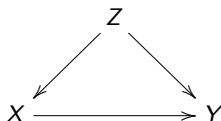
- Most epidemiological research questions are concerned with causality
- Typically, we want to know if the exposure has a causal effect on the outcome
  - hypothesis testing
- If there is a causal effect, then we typically also want to know how big it is
  - effect estimation



- In ideal randomized trials we have exchangeability:

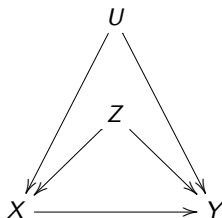
$$(Y_0, Y_1) \perp\!\!\!\perp X$$

- As a consequence, association = causation



- In observational studies we usually don't have exchangeability, because of confounding
- We may achieve conditional exchangeability by controlling for an appropriate set of covariates:

$$(Y_0, Y_1) \perp\!\!\!\perp X \mid Z$$



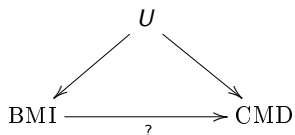
- We can control for confounders that we have measured
- But unmeasured confounders may still bias the results

- Instrumental variable (IV) methods is a collection of methods with a remarkable property
- **They can be used to test for/estimate causal effects in the presence of unmeasured confounding**
- IV methods have a long history in econometrics
- They have more recently become popular in epidemiology
  - in particular through Mendelian randomization (MR) studies, where the IV is a (set of) gene(s)

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- Kivimäki et al. (2011) aimed to study if obesity has a causal effect on common mental disorders (CMD)
- Obesity was measured with body mass index (BMI)
- CMD was measured with a continuous score, subsequently dichotomized into 'high' and 'low'
- BMI and CMD were measured on 4 occasions during follow-up
- The exposure was defined as mean BMI across the 4 occasions
- The outcome was defined as the number of times, out of the 4 occasions, that the subject had a high CMD score
  - integer, ranging from 0 to 4





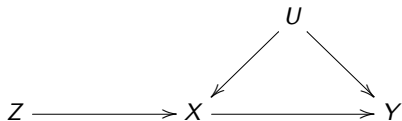
- Large potential for confounding, e.g. by
  - age
  - sex
  - socioeconomic status (SES)
  - comorbid conditions
  - etc

- The study participants were genotyped on the fat mass and obesity (FTO) gene
- It has previously been shown that this gene affects BMI
- Kivimäki et al. (2011) used the FTO gene as an IV in the analysis
  - a Mendelian randomization (MR) study

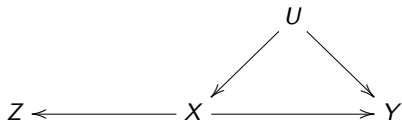
- IV methods can be used to test for/estimate causal effects in the presence of unmeasured confounding
- No free lunch - all IV methods rely on strong assumptions:
  - **1. The IV is associated with the exposure (relevance)**
  - **2. The IV may only affect the outcome through the exposure (exclusion)**
  - **3. The association between the IV and the outcome is unconfounded (exogeneity)**
- (Alternative formulations occur)
- *Display the IV assumptions in a DAG*

- 1. The IV is associated with the exposure

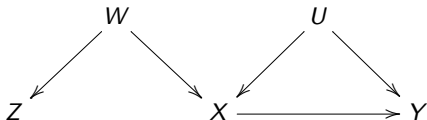
- e.g. the IV affects the exposure:



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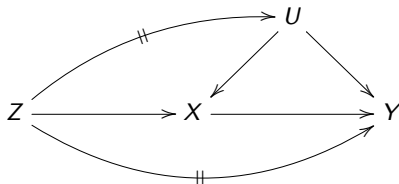


- e.g. the IV and the exposure have common causes:

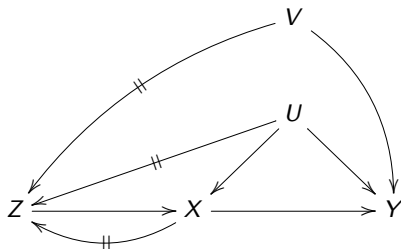


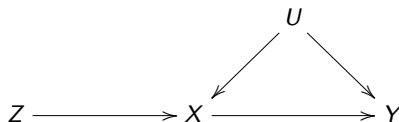
- For convenience we will only consider the upper DAG

- 2. The IV may only affect the outcome through the exposure

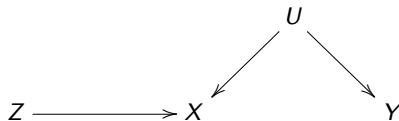


- 3. The association between the IV and the outcome is unconfounded



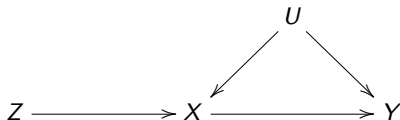


- Suppose that we observe a statistical association between  $Z$  and  $Y$  in the DAG above
- *Is it then possible that  $X$  has no causal effect on  $Y$ ?*

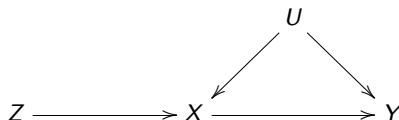


- No. If  $X$  has no causal effect on  $Y$ , then there is no open path between  $Z$  and  $Y$
- Hence, if  $X$  has no causal effect on  $Y$ , then  $Z$  and  $Y$  should be independent





- And the other way around: **if  $Z$  and  $Y$  are associated, then  $X$  has a causal effect on  $Y$**
- This suggests a simple method for hypothesis testing
  - check if  $Z$  is associated with  $Y$
  - if it is, then conclude that  $X$  has a causal effect on  $Y$



- This IV hypothesis test is general in that it allows for
  - arbitrary types of variables (binary, categorical, continuous, time-to-event etc)
  - arbitrary statistical tests (non-parametric test,  $\chi^2$ -test,  $t$ -test, regression model etc)
  - arbitrary sampling schemes (cross-sectional, prospective, retrospective etc)
- ... as long as the three IV assumptions hold

# Results from the Kivimäki et al. (2011) study

**Table 2.** Associations of Fat Mass and Obesity-Associated (*FTO*) Genotype With Body Mass Index, Overweight, Obesity, and Common Mental Disorders From 4 Repeated Assessments in Men and Women, Whitehall II Study, 1985–2004

| Predictor                           | No. of Participants | BMI <sup>a</sup> |               | Overweight <sup>b</sup> |               | Obesity <sup>b</sup> |               | Common Mental Disorders <sup>c</sup> |               |
|-------------------------------------|---------------------|------------------|---------------|-------------------------|---------------|----------------------|---------------|--------------------------------------|---------------|
|                                     |                     | β                | 95% CI        | β                       | 95% CI        | β                    | 95% CI        | β                                    | 95% CI        |
| Men                                 |                     |                  |               |                         |               |                      |               |                                      |               |
| No. of <i>FTO</i> adiposity alleles |                     |                  |               |                         |               |                      |               |                                      |               |
| 0                                   | 1,046               | 0.00             | Referent      | 0.00                    | Referent      | 0.00                 | Referent      | 0.00                                 | Referent      |
| 1                                   | 1,442               | 0.283            | 0.039, 0.526  | 0.200                   | 0.064, 0.335  | 0.018                | −0.054, 0.090 | 0.020                                | −0.065, 0.105 |
| 2                                   | 493                 | 0.935            | 0.608, 1.262  | 0.331                   | 0.144, 0.519  | 0.192                | 0.094, 0.289  | 0.172                                | 0.058, 0.286  |
| Per-allele increase                 | 2,981               | 0.433            | 0.275, 0.592  | 0.173                   | 0.083, 0.262  | 0.081                | 0.034, 0.129  | 0.074                                | 0.019, 0.129  |
| <i>P</i> for trend                  |                     | <0.0001          |               | <0.0001                 |               | 0.001                |               | 0.009                                |               |
| Women                               |                     |                  |               |                         |               |                      |               |                                      |               |
| No. of <i>FTO</i> adiposity alleles |                     |                  |               |                         |               |                      |               |                                      |               |
| 0                                   | 448                 | 0.00             | Referent      | 0.00                    | Referent      | 0.00                 | Referent      | 0.00                                 | Referent      |
| 1                                   | 543                 | −0.183           | −0.741, 0.376 | 0.015                   | −0.196, 0.225 | −0.076               | −0.441, 0.289 | 0.073                                | −0.074, 0.219 |
| 2                                   | 173                 | 0.310            | −0.474, 1.094 | 0.145                   | −0.155, 0.445 | 0.121                | −0.391, 0.632 | −0.011                               | −0.217, 0.194 |
| Per-allele increase                 | 1,164               | 0.077            | −0.294, 0.449 | 0.058                   | −0.083, 0.120 | 0.012                | −0.089, 0.114 | 0.012                                | −0.085, 0.110 |
| <i>P</i> for trend                  |                     | 0.68             |               | 0.42                    |               | 0.81                 |               | 0.80                                 |               |

Abbreviations: BMI, body mass index; CI, confidence interval; *FTO*, fat mass and obesity-associated gene; GHQ, General Health Questionnaire.

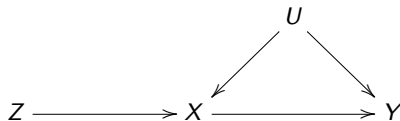
<sup>a</sup> Mean BMI (weight (kg)/height (m)<sup>2</sup>) across 4 repeated clinical examinations conducted over a 19-year follow-up period.

<sup>b</sup> Number of times (range, 0–4) a participant was found to be overweight (BMI 25.0–29.9) or obese (BMI  $\geq 30$ ) in 4 examinations conducted over a 19-year follow-up period. The analysis of overweight did not include obese participants and was therefore based on 2,473 men and 861 women.

<sup>c</sup> Number of times (range, 0–4) a participant was designated a GHQ “case” in 4 examinations conducted over a 19-year follow-up period. GHQ score was used as the measure of common mental disorders.

- The exposure is BMI, not the FTO gene
- If we believe that the FTO gene is a valid IV, then the results indicate a causal effect of BMI on CMD for men
  - no evidence for a causal effect for women

- 1. The IV is associated with the exposure



- Easy to verify

# Results from the Kivimäki et al. (2011) study

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|                                     |                     | β                | 95% CI        | β                       | 95% CI        | β                    | 95% CI        | β                                    | 95% CI        |
| <i>Men</i>                          |                     |                  |               |                         |               |                      |               |                                      |               |
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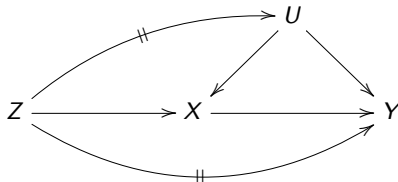
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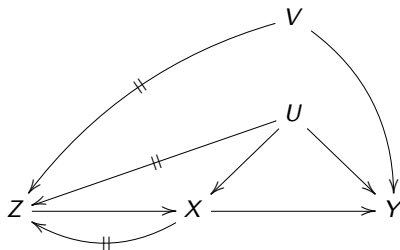
<sup>c</sup> Number of times (range, 0–4) a participant was designated a GHQ “case” in 4 examinations conducted over a 19-year follow-up period. GHQ score was used as the measure of common mental disorders.

- 2. The IV may only affect the outcome through the exposure



- Hard to verify

- 3. The association between the IV and the outcome is unconfounded



- Hard to verify



# Results from the Kivimäki et al. (2011) study

**Table 3.** Associations of Fat Mass and Obesity-Associated (*FTO*) Genotype With Factors Potentially Confounding the Relation Between Obesity and Common Mental Disorders in Men and Women, Whitehall II Study, 1985–2004

| Baseline<br>Characteristic            | No. of <i>FTO</i> Adiposity Alleles |      |            |       |      |            |     |      |            | <i>P</i> Value |
|---------------------------------------|-------------------------------------|------|------------|-------|------|------------|-----|------|------------|----------------|
|                                       | 0                                   |      |            | 1     |      |            | 2   |      |            |                |
|                                       | No.                                 | %    | Mean (SD)  | No.   | %    | Mean (SD)  | No. | %    | Mean (SD)  |                |
| <i>Men</i>                            |                                     |      |            |       |      |            |     |      |            |                |
| Maximum no. of participants           | 1,046                               |      |            | 1,442 |      |            | 493 |      |            |                |
| Age, years                            |                                     |      | 44.0 (6.1) |       |      | 43.7 (5.9) |     |      | 44.0 (5.7) | 0.75           |
| Marital status (married)              |                                     | 82.8 |            |       | 84.4 |            |     | 82.4 |            | 0.45           |
| Socioeconomic position                |                                     |      |            |       |      |            |     |      |            |                |
| High                                  |                                     | 39.6 |            |       | 41.5 |            |     | 44.4 |            | 0.39           |
| Intermediate                          |                                     | 54.8 |            |       | 53.7 |            |     | 51.3 |            |                |
| Low                                   |                                     | 5.6  |            |       | 4.9  |            |     | 4.3  |            |                |
| Chronic disease                       |                                     | 8.3  |            |       | 10.8 |            |     | 9.7  |            | 0.12           |
| Prevalent coronary heart disease      |                                     | 0.8  |            |       | 0.6  |            |     | 0.8  |            | 0.75           |
| Prevalent diabetes                    |                                     | 0.5  |            |       | 0.6  |            |     | 0.8  |            | 0.72           |
| Use of psychotropic drugs             |                                     | 2.3  |            |       | 2.0  |            |     | 2.2  |            | 0.88           |
| High alcohol consumption <sup>a</sup> |                                     | 16.4 |            |       | 17.4 |            |     | 20.6 |            | 0.13           |
| Physical inactivity                   |                                     | 23.8 |            |       | 19.8 |            |     | 22.6 |            | 0.051          |
| Current smoking                       |                                     | 10.4 |            |       | 11.9 |            |     | 9.2  |            | 0.20           |

# Results from the Kivimäki et al. (2011) study, cont'd

|                                       | Women      |      |            |                 |
|---------------------------------------|------------|------|------------|-----------------|
| Maximum no. of participants           | 448        | 543  | 173        |                 |
| Age, years                            | 44.0 (6.1) |      | 44.5 (5.8) | 44.7 (5.9) 0.13 |
| Marital status (married)              | 60.5       | 57.3 | 61.9       | 0.44            |
| Socioeconomic position                |            |      |            |                 |
| High                                  | 15.2       | 16.1 | 19.7       | 0.20            |
| Intermediate                          | 44.0       | 46.5 | 49.7       |                 |
| Low                                   | 40.9       | 37.5 | 30.6       |                 |
| Chronic disease                       | 10.4       | 9.2  | 8.8        | 0.78            |
| Prevalent coronary heart disease      | 0.4        | 0.4  | 0.0        | 0.69            |
| Prevalent diabetes                    | 0.5        | 0.4  | 0.0        | 0.69            |
| Use of psychotropic drugs             | 4.0        | 4.4  | 4.1        | 0.94            |
| High alcohol consumption <sup>b</sup> | 3.0        | 3.5  | 5.8        | 0.24            |
| Physical inactivity                   | 41.3       | 38.0 | 39.3       | 0.59            |
| Current smoking                       | 13.3       | 17.3 | 17.1       | 0.20            |

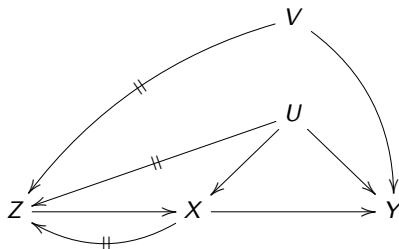
Abbreviations: *FTO*, fat mass and obesity-associated gene; SD, standard deviation.

<sup>a</sup> >21 alcohol units per week.

<sup>b</sup> >14 alcohol units per week.

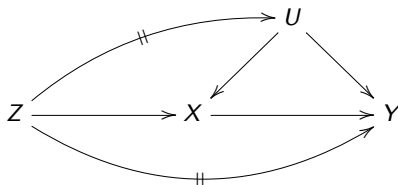
- In the Kivimäki et al. (2011) study, the IV appears is independent of all (or most) measured covariates
- *Which of the IV assumption does this observation support?*

- 3. The association between the IV and the outcome is unconfounded



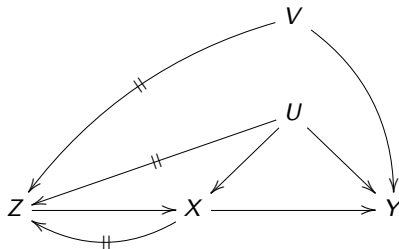
- All measured covariates may affect the outcome
- However, we know *a priori* that none of the measured covariates can affect the IV
  - genes are not affected by age, marital and socioeconomic status, comorbid conditions, use of drugs and alcohol, physical inactivity or smoking
- Hence, we know *a priori* that none of the measured covariates can violate the third assumption

- 2. The IV may only affect the outcome through the exposure

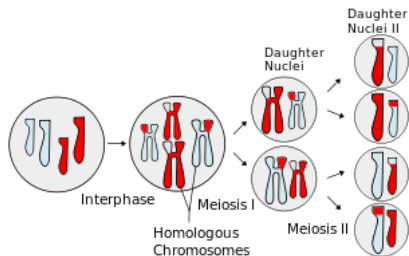


- A priori*, the IV may affect most of the measured covariates
  - genes may affect marital and socioeconomic status, comorbid conditions, use of drugs and alcohol, physical inactivity and smoking
- If any of these covariates has a direct (not through the exposure) effect on the outcome, then the second assumption is violated
- However, the lack of association between the IV and the covariates indicates no effect of the IV on the covariates
- Hence, this gives some support to the second assumption

- 3. The association between the IV and the outcome is unconfounded

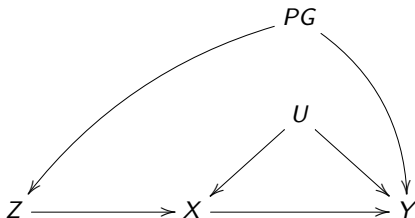


- If genes were randomized, then this assumption would always be true in MR studies
- Are genes randomized?



- In meiosis, the child randomly receives one allele from the father and one allele from the mother
- In this sense, the child's alleles are randomized **conditional on the parents' genes**
  - thus the term 'Mendelian randomization'
- However, in MR studies we typically don't genotype the parents of the study participants
  - cannot condition on the parents' genes in the analysis
- Not correct to view the study participant genes as randomized

- The parents' genes clearly affect the child's genes
- Suppose that the parents' genes also have a direct (not through the exposure) effect on the child's outcome
  - e.g. parents' genes may affect how they treat the child, which in turn affects the child's risk of mental disorders

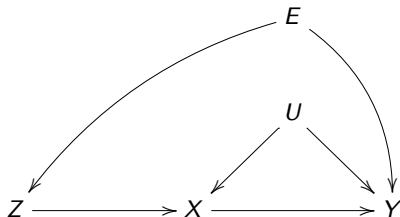


- Then assumption 3 would be violated



# Population mixture

- Suppose that the study population consists of different ethnic groups, with different distributions of the genetic IV
- Suppose that ethnicity also has a direct (not through the exposure) effect on the outcome
  - e.g. due to social or cultural differences, some ethnic groups may be more likely to develop mental disorders

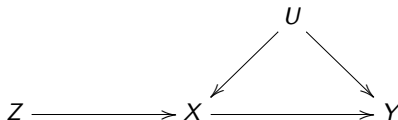


- Then assumption 3 would be violated

- IV methods can be used to test for/estimate causal effects in the presence of unmeasured confounding
- No free lunch - all IV methods rely on strong assumptions
- These assumptions are to a large extent untestable
  - need subject matter knowledge to determine whether the assumptions are plausible on case-by-case basis

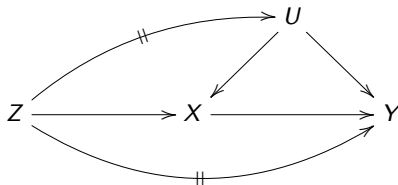
- $Z$  = randomly assigned treatment
  - e.g. new treatment vs old treatment
- $X$  = treatment actually taken
- $Y$  = outcome of interest
- Want the causal effect of  $X$  on  $Y$
- $U$  = all factors that affect both the outcome and the decision to take the new or old treatment
  - e.g. younger and healthier subjects may be more inclined to take the new treatment
- **The assigned treatment can be used as an IV**

- 1. The IV is associated with the exposure



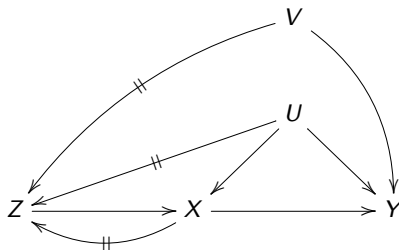
- This assumption holds by design, since the treatment assignment affects the decision to take the new or old treatment

- 2. The IV may only affect the outcome through the exposure

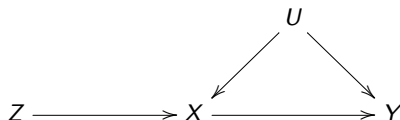


- A direct (not through the exposure) effect of the treatment assignment is possible in principle, but often unlikely in practice

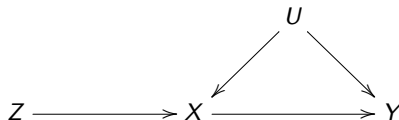
- 3. The association between the IV and the outcome is unconfounded



- This assumption holds by design, since the treatment assignment is randomized



- One possible method to analyze randomized trials with non-compliance is the 'as-treated' (AT) analysis
- In this analysis, one tests for an association between the treatment actually taken ( $X$ ) and the outcome ( $Y$ )
- This analysis is typically confounded (by  $U$ ), and does not give a causal effect



- The most common method to analyze randomized trials with non-compliance is the 'intention-to-treat' (ITT) analysis
- In this analysis, one tests for an association between the assigned treatment ( $Z$ ) and the outcome ( $Y$ )
- This is exactly the IV hypothesis test we discussed previously
- The ITT analysis proves whether the treatment actually taken has a causal effect on the outcome



- Sommer and Zeger (1991) considered a randomized trial of vitamin A supplement in northern Sumatra
- At baseline, 24492 preschool children were randomized to treatment or no treatment with vitamin A
- All deaths were ascertained during a follow-up of 12 months

## Results from the Sommer and Zeger (1991) study

Table I. Mortality rates in control and programme villages, months 4-12, stratified by compliance

| Study group | Complied | Children | Deaths | Mortality (per 1000) |
|-------------|----------|----------|--------|----------------------|
| Control     | —        | 11,588   | 74     | 6.4                  |
| Treatment   | —        | 12,094   | 46     | 3.8                  |
|             | Yes      | 9,675    | 12     | 1.2                  |
|             | No       | 2,419    | 34     | 14.1                 |

An intent-to-treat analysis was performed to estimate the programmatic effectiveness of the vitamin A distribution. The mortality in the two groups from month 4 (following completion of the first distribution cycle) to month 12 was compared. As shown in Table I, there were 46 deaths out of 12,094 children (0.38 per cent) in the villages randomized to vitamin A, as opposed to 74 out of 11,588 (0.64 per cent) in control villages. The effectiveness relative risk is estimated by  $0.38/0.64 = 0.59$  (95 per cent confidence interval: 0.41 to 0.86). Hence, the data suggest roughly a 40 per cent reduction in mortality associated with the vitamin A distribution programme.

- This the effect of 'treatment assignment' (the ITT effect), not the effect of treatment
- What is the treatment effect?

- We have seen that we can **test** for a causal exposure effect, assuming that the IV assumptions hold
- Balke and Pearl (1997) showed that the IV assumptions are not enough to **estimate** the causal exposure effect
- However, they showed that the causal effect can be bounded
  - i.e. that it is possible to give an interval that includes the true value of the effect

- Let  $Y$ ,  $X$  and  $Z$  be binary
- Define

$$p_{YX.Z} = p(Y, X|Z)$$

and

$$\psi = p(Y_1 = 1) - p(Y_0 = 1)$$

- Balke and Pearl (1997) showed that  $\psi$  is bounded by

$$\max \left\{ \begin{array}{l} p_{00.0} + p_{11.1} - 1 \\ p_{00.1} + p_{11.1} - 1 \\ p_{11.0} + p_{00.1} - 1 \\ p_{00.0} + p_{11.0} - 1 \\ 2p_{00.0} + p_{11.0} + p_{10.1} + p_{11.1} - 2 \\ p_{00.0} + 2p_{11.0} + p_{00.1} + p_{01.1} - 2 \\ p_{10.0} + p_{11.0} + 2p_{00.1} + p_{11.1} - 2 \\ p_{00.0} + p_{01.0} + p_{00.1} + 2p_{11.1} - 2 \end{array} \right\} \leq \psi \leq \min \left\{ \begin{array}{l} 1 - p_{10.0} - p_{01.1} \\ 1 - p_{01.0} - p_{10.1} \\ 1 - p_{01.0} - p_{10.0} \\ 1 - p_{01.1} - p_{10.1} \\ 2 - 2p_{01.0} - p_{10.0} - p_{10.1} - p_{11.1} \\ 2 - p_{01.0} - 2p_{10.0} - p_{00.1} - p_{01.1} \\ 2 - p_{10.0} - p_{11.0} - 2p_{01.1} - p_{10.1} \\ 2 - p_{00.0} - p_{01.0} - p_{01.1} - 2p_{10.1} \end{array} \right\}$$

# The Sommer and Zeger (1991) study revisited

Table I. Mortality rates in control and programme villages, months 4–12, stratified by compliance

| Study group | Complied | Children | Deaths | Mortality (per 1000) |
|-------------|----------|----------|--------|----------------------|
| Control     | —        | 11,588   | 74     | 6.4                  |
| Treatment   | —        | 12,094   | 46     | 3.8                  |
|             | Yes      | 9,675    | 12     | 1.2                  |
|             | No       | 2,419    | 34     | 14.1                 |

- $Z = 1$  for 'assigned to vitamin A',  $X = 1$  for 'took vitamin A',  $Y = 1$  for 'survived'
- To compute the bounds for the causal risk difference we need  $p_{YX,Z}$  for all combinations of  $Y$ ,  $X$  and  $Z$

$$p_{00.0} = 74/11588 = 0.0064$$

$$p_{01.0} = 0/11588 = 0$$

$$p_{10.0} = (11588 - 74)/11588 = 0.9936$$

$$p_{11.0} = 0/11588 = 0$$

$$p_{00.1} = 34/12094 = 0.0028$$

$$p_{01.1} = 12/12094 = 0.0010$$

$$p_{10.1} = (2419 - 34)/12094 = 0.1972$$

$$p_{11.1} = (9675 - 12)/12094 = 0.7990$$

plug in the computed results in the formula from the previous slide to find the bounds.

$$-0.1946 \leq \psi \leq 0.0054$$

- Given the observed data, the causal risk difference can be as small as -0.1946 and as large as 0.0054
- The chance of survival may be anywhere between
  - 19.46 percentage-points lower, and
  - 0.54 percentage-points higher

if everybody were treated with vitamin A, as compared to if nobody were treated with vitamin A

$$-0.1946 \leq \psi \leq 0.0054$$

- The bounds include the value 0, thus it is possible that vitamin A has no effect on survival
- But we observed an ITT effect! Doesn't this prove that vitamin A has an effect on survival?
- The bounds include negative values, thus it is possible that vitamin A has a negative effect on survival
- But the observed ITT effect was positive! Doesn't this prove that vitamin A has a positive effect on survival?

- The presence of an ITT effect implies that there is a causal treatment effect **for some subjects in the population**
- It does not imply that
  - there is a causal treatment effect for all subjects in the population
  - the causal treatment effect has the same direction for all subjects in the population
- The population causal effect may be 0, or negative, if there is strong heterogeneity in treatment response
  - see Balke and Pearl (1997) for numerical example



$$-0.1946 \leq \psi \leq 0.0054$$

- The bounds should not be confused with a confidence interval
- A confidence interval quantifies the uncertainty due to sampling variability
  - decreases with sample size
- The bounds quantify the uncertainty due to unmeasured confounding
  - does not decrease with sample size
- In practice; estimate the lower and upper bound, and compute confidence intervals for the estimates

- 1 IV - Introduction
- 2 Origins and overview
- 3 Hypothesis testing
- 4 Verifying the IV assumptions
- 5 The randomized trial with non-compliance
- 6 Effect estimation
  - The Two-Stage estimation method

- In many studies, the obtained bounds are very wide and contain 0 as a possible causal effect. Namely, indicate how much (or little!) we can say about the causal effect, using only the IV assumptions
- But sometimes additional assumptions are reasonable, such as (approximate) homogeneity in treatment response
- Also, we often want to consider non-binary variables
- These extensions naturally lead to regression modeling

$$E(Y_x|U) = \psi x + f(U)$$

- $f(U)$  is an unspecified (linear or non-linear) function of  $U$

$$E(Y_x|U) = \psi x + f(U)$$

- $\psi$  is the conditional causal effect of  $X$  on  $Y$ , given  $U$
- It measures the increase in the mean of  $Y$ , when  $X$  is increased with 1 unit, in the subpopulation of subjects with a given level of  $U$

$$\begin{aligned} E(Y_{x+1}|U) - E(Y_x|U) \\ &= \{\psi(x+1) + f(U)\} - \{\psi x + f(U)\} \\ &= \psi \end{aligned}$$

$$E(Y_x|U) = \psi x + f(U)$$

- Linear conditional effect of  $X$ , given  $U$
- The conditional effect of  $X$ , given  $U$ , is the same across levels of  $U$ 
  - no effect modification by  $U$
  - aka no interaction between  $X$  and  $U$  on the linear scale
  - aka homogeneity in treatment response

$$E(Y_x|U) = \psi x + f(U)$$

- The conditional effect of  $X$ , given  $U$ , is the same across levels of  $U$
- The mean difference is collapsible; the constant conditional effect  $\psi$  is equal to the marginal effect
- It measures the increase in the mean of  $Y$ , when  $X$  is increased with 1 unit, in the whole population

$$E(Y_x|U) = \psi x + f(U)$$

- A common method to estimate  $\psi$  is 'two-stage' (TS) estimation
- First stage: fit a linear regression model for  $X$  on  $Z$
- Second stage: fit a linear regression model for  $Y$  on  $Z$
- Estimate  $\psi$  as the ratio of the two regression slopes (second stage/first stage)
- TS-estimation gives an unbiased estimate of  $\psi$  **even if the first and second stage models are incorrect**, provided that
  - the IV assumptions hold, and
  - the causal linear model is correct



- First stage: fit a linear regression model for  $X$  on  $Z$
- Create predictions  $\hat{X}$  from the first stage model
- Second stage: fit a linear regression model for  $Y$  on  $\hat{X}$
- The estimate of  $\psi$  is the coefficient for  $\hat{X}$  in the second stage model
- The two formulations give identical estimates of  $\psi$  when all models are linear

# TS-estimation for the Kivimäki et al. (2011) study

**Table 2.** Associations of Fat Mass and Obesity-Associated (*FTO*) Genotype With Body Mass Index, Overweight, Obesity, and Common Mental Disorders From 4 Repeated Assessments in Men and Women, Whitehall II Study, 1985–2004

| Predictor                           | No. of Participants | BMI <sup>a</sup> |              | Overweight <sup>b</sup> |              | Obesity <sup>b</sup> |               | Common Mental Disorders <sup>c</sup> |               |
|-------------------------------------|---------------------|------------------|--------------|-------------------------|--------------|----------------------|---------------|--------------------------------------|---------------|
|                                     |                     | β                | 95% CI       | β                       | 95% CI       | β                    | 95% CI        | β                                    | 95% CI        |
| Men                                 |                     |                  |              |                         |              |                      |               |                                      |               |
| No. of <i>FTO</i> adiposity alleles |                     |                  |              |                         |              |                      |               |                                      |               |
| 0                                   | 1,046               | 0.00             | Referent     | 0.00                    | Referent     | 0.00                 | Referent      | 0.00                                 | Referent      |
| 1                                   | 1,442               | 0.283            | 0.039, 0.526 | 0.200                   | 0.064, 0.335 | 0.018                | −0.054, 0.090 | 0.020                                | −0.065, 0.105 |
| 2                                   | 493                 | 0.935            | 0.608, 1.262 | 0.331                   | 0.144, 0.519 | 0.192                | 0.094, 0.289  | 0.172                                | 0.058, 0.286  |
| Per-allele increase                 | 2,981               | 0.433            | 0.275, 0.592 | 0.173                   | 0.083, 0.262 | 0.081                | 0.034, 0.129  | 0.074                                | 0.019, 0.129  |
| <i>P</i> for trend                  |                     | <0.0001          |              | <0.0001                 |              | 0.001                |               | 0.009                                |               |

$$\hat{\psi}_{men} = 0.074/0.433 = 0.17$$

# TS-estimation for the Kivimäki et al. (2011) study, cont'd

| Women                               |       |        |               |       |               |        |               |        |               |
|-------------------------------------|-------|--------|---------------|-------|---------------|--------|---------------|--------|---------------|
| No. of <i>FTO</i> adiposity alleles |       |        |               |       |               |        |               |        |               |
| 0                                   | 448   | 0.00   | Referent      | 0.00  | Referent      | 0.00   | Referent      | 0.00   | Referent      |
| 1                                   | 543   | -0.183 | -0.741, 0.376 | 0.015 | -0.196, 0.225 | -0.076 | -0.441, 0.289 | 0.073  | -0.074, 0.219 |
| 2                                   | 173   | 0.310  | -0.474, 1.094 | 0.145 | -0.155, 0.445 | 0.121  | -0.391, 0.632 | -0.011 | -0.217, 0.194 |
| Per-allele increase                 | 1,164 | 0.077  | -0.294, 0.449 | 0.058 | -0.083, 0.120 | 0.012  | -0.089, 0.114 | 0.012  | -0.085, 0.110 |
| <i>P</i> for trend                  |       | 0.68   |               | 0.42  |               | 0.81   |               | 0.80   |               |

Abbreviations: BMI, body mass index; CI, confidence interval; *FTO*, fat mass and obesity-associated gene; GHQ, General Health Questionnaire.

<sup>a</sup> Mean BMI (weight (kg)/height (m)<sup>2</sup>) across 4 repeated clinical examinations conducted over a 19-year follow-up period.

<sup>b</sup> Number of times (range, 0–4) a participant was found to be overweight (BMI 25.0–29.9) or obese (BMI ≥30) in 4 examinations conducted over a 19-year follow-up period. The analysis of overweight did not include obese participants and was therefore based on 2,473 men and 861 women.

<sup>c</sup> Number of times (range, 0–4) a participant was designated a GHQ “case” in 4 examinations conducted over a 19-year follow-up period. GHQ score was used as the measure of common mental disorders.

$$\hat{\psi}_{women} = 0.012/0.077 = 0.16$$

- When  $X$  and  $Y$  are binary,  $\psi$  is the causal risk difference

$$p(Y_1 = 1|U) - p(Y_0 = 1|U) = p(Y_1 = 1) - p(Y_0 = 1)$$

- When  $Z$  is binary as well, the TS estimate of  $\psi$  is

$$\frac{p(Y = 1|Z = 1) - p(Y = 1|Z = 0)}{p(X = 1|Z = 1) - p(X = 1|Z = 0)}$$

- The numerator is the ITT effect, as a risk difference
- The denominator is equal to 1 if 'compliance' is 100%

Table I. Mortality rates in control and programme villages, months 4-12, stratified by compliance

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|             | Yes      | 9,675    | 12     | 1.2                  |
|             | No       | 2,419    | 34     | 14.1                 |

- $Z = 1$  for 'assigned to vitamin A',  $X = 1$  for 'took vitamin A',  $Y = 1$  for 'survived'

$$p(Y = 1|Z = 0) = (11588 - 74)/11588 = 0.9936$$

$$p(Y = 1|Z = 1) = (12094 - 12 - 34)/12094 = 0.9962$$

$$p(X = 1|Z = 0) = 0/11588 = 0$$

$$p(X = 1|Z = 1) = 9675/12094 = 0.8000$$

$$\hat{\psi} = \frac{p(Y = 1|Z = 1) - p(Y = 1|Z = 0)}{p(X = 1|Z = 1) - p(X = 1|Z = 0)} = 0.0032$$

- Bounds (Balke, A. and Pearl, J., 1997):

$$-0.1946 \leq \psi \leq 0.0054$$

- TS-estimation:

$$\hat{\psi} = 0.0032$$

- Clearly, the TS estimate relies heavily on the assumption of homogeneity in treatment response

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```
> vitA <- data.frame(Z=c(0,0,1,1,1,1),
X=c(0,0,0,0,1,1), Y=c(0,1,0,1,0,1))
> vitA
  Z X Y
1 0 0 0
2 0 0 1
3 1 0 0
4 1 0 1
5 1 1 0
6 1 1 1
> vitA <- vitA[c(rep(1,74), rep(2,11588-74), rep(3,34),
rep(4,2419-34), rep(5,12), rep(6,9675-12)), ]
```

```
> library(ivtools)
> fitX.Z <- glm(formula=X~Z, data=vitA)
> fitY.X <- glm(formula=Y~X, data=vitA)
> fitIV <- ivglm(estmethod="ts", fitX.LZ=fitX.Z,
fitY.LX=fitY.X, data=vitA)
> summary(fitIV)
```

```
Estimate Std. Error  z value Pr(>|z|)
(Intercept) 0.993614    0.000740 1342.741 < 2e-16 ***
X            0.003228    0.001159   2.785  0.00536 **
```

```
> confint(fitIV)
2.5 %      97.5 %
(Intercept) 0.9921637313 0.995064436
X            0.0009560731 0.005500004
```

- Sjölander and Martinussen (2019)



- TS-estimation has the advantage of being simple
- However, for non-linear (e.g. logistic, Cox proportional hazards) regression models it gives biased estimates
  - at best the bias is small, but the bias can be substantial
- A more sophisticated method - G-estimation - gives unbiased estimates in non-linear models
  - Vansteelandt et al. (2011), Martinussen et al. (2018)
  - also available in `ivtools`

- IV methods can be used to test for/estimate causal effects in the presence of unmeasured confounding
- All IV methods rely on the IV assumptions
- The IV assumptions are sufficient for hypothesis testing
- Effect estimation requires parametric model assumptions as well
  - e.g. homogeneity in treatment response
- The two-stage procedure assume linear causal model: The causal effect can be estimated with two-stage estimation
  - unbiased estimate in linear models, but potentially biased in non-linear models
- G-estimation is more complex, but generally unbiased

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